Next-order in Rossby SQG^{+1} model for reconstructing surface vorticity from SSH

Ryan Shìjié Dù[1] (ryan_sjdu@nyu.edu), Shafer Smith[1], Dhruv Balwada[2], Spencer Jones[3]

[1]New York University, [2]Lamont-Doherty Earth Observatory, [3]Texas A&M University College Station

eSQG^{+1}

- eSQG diagnoses velocities from SSHA (via geostrophic balance) and projects them into the interior [4]

$$\hat{\Phi}^0(k,\ell,z) = \frac{g}{f}\hat{\eta}^{\text{data}}(k,\ell)\exp\left(\frac{N}{f}Kz\right).$$

- eSQG^{+1} extends eSQG, accounting for higher-order in Rossby terms in the velocity reconstruction through the QG^{+1} asymptotic model [3, 2].
- The eSQG^{+1} vorticity is

$$\zeta = \nabla^2\Phi^0 + \{\text{Ro}\}(\nabla^2\Phi^1 + F^1_{yz} - G^1_{xz}).$$

where Φ^0 is determined by its surface value $\Phi^{0,s}$, and the potentials F^1, G^1 and Φ^1 are nonlinear functions of Φ^0.

- Pressure relates to vorticity by cyclogeostrophic balance

$$\nabla^2\eta^{\text{model}} - f\zeta/g = \{\text{Ro}\}2J(\Phi^0_x, \Phi^0_y)/g.$$

Nonlinear inverse problem

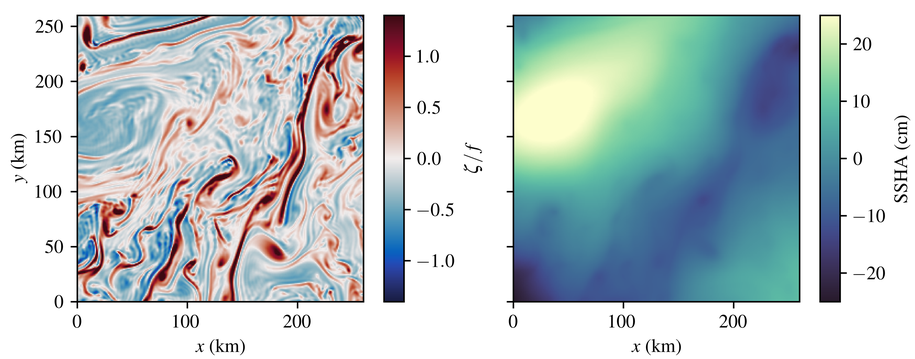
- The eSQG^{+1} model SSHA η^{model} is a complex nonlinear function of $\Phi^{0,s}$.
- eSQG^{+1} reconstruction seek to solve for $\Phi^{0,s}$ that match well with a given SSHA observation (η^{data}), by minimizing

$$\min \left\|\eta^{\text{data}} - \eta^{\text{model}}(\Phi^{0,s})\right\|_{H^1}$$

- Minimization is done using Limited-memory BFGS, a gradient based, quasi-Newton minimization algorithm [5].
- Differentiation through SQG^{+1} is done automatically using JAX.
- Note some low-pass filter is applied to the SSHA field and the reconstructed eSQG^{+1} vorticity.

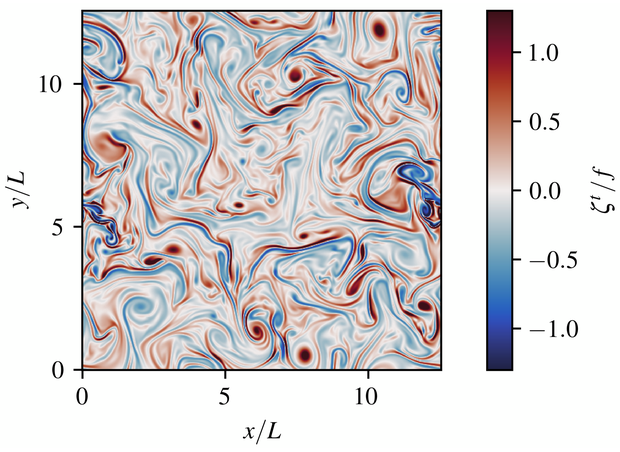
Target of reconstruction

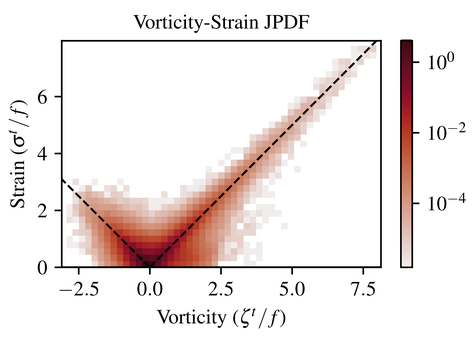
- We use the primitive equation, steadily-forced channel simulation of [1] as the target, and attempt to reconstruct surface vorticity.
- The simulation exhibits minimal gravity wave motions, simplifying the reconstruction task.



QG^{+1} captures submesoscale

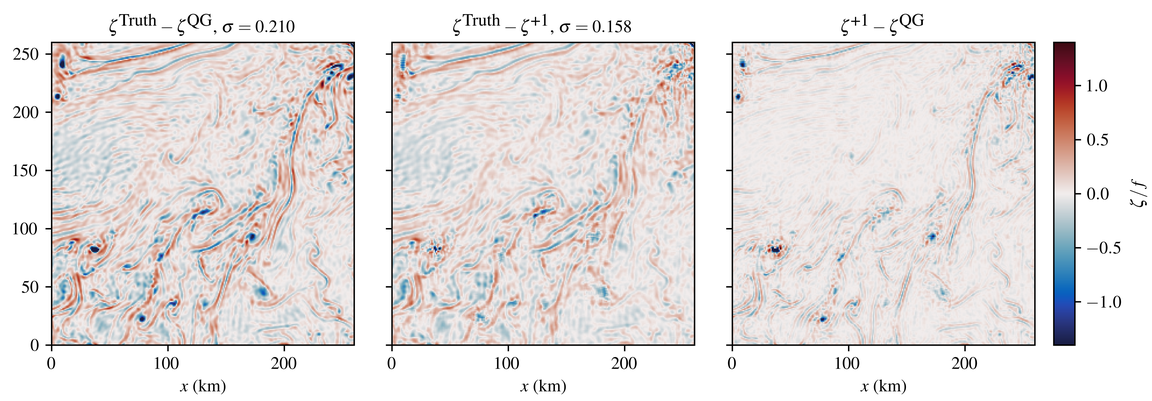
- As a forward model, the evolution of QG^{+1} exhibits true frontogenesis and captures realistic statistics of the balanced component of the submesoscale.



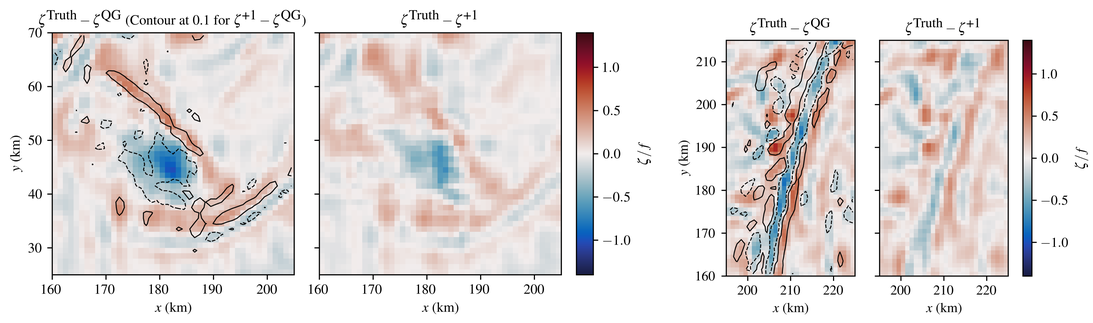


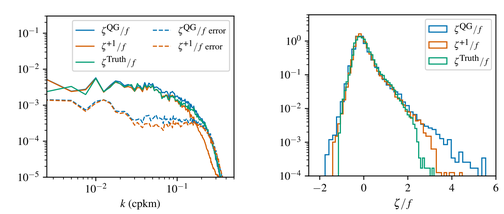
Analysis of the performance

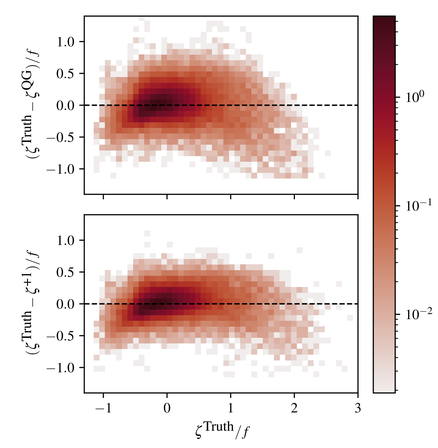
- eSQG^{+1} leads to overall decrease of the error (25%), particularly at the small scale $O(10)$ km.
- eSQG^{+1} improves the regions of strong cyclonic small vortices and fronts.
- Geostrophic balance overexaggerated strong cyclones. eSQG^{+1} correct this.



Zoomed in view of an eddy and a front.







Preprint [2]

References

[1] D. Balwada, K. S. Smith, and R. Abernathey. Submesoscale Vertical Velocities Enhance Tracer Subduction in an Idealized Antarctic Circumpolar Current. *Geophysical Research Letters*, 45(18):9790–9802, 2018.

[2] R. S. Dù, K. S. Smith, and O. Bühler. Next-order balanced model captures submesoscale physics and statistics. *arXiv*, 2408.03422, Submittd to *Journal of Physical Oceanography*.

[3] G. J. Hakim, C. Snyder, and D. J. Muraki. A New Surface Model for Cyclone–Anticyclone Asymmetry. *Journal of the Atmospheric Sciences*, 59(16):2405–2420, Aug. 2002.

[4] G. Lapeyre and P. Klein. Dynamics of the Upper Oceanic Layers in Terms of Surface Quasigeostrophy Theory. *Journal of Physical Oceanography*, 36(2):165–176, Feb. 2006.

[5] D. C. Liu and J. Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45(1):503–528, Aug. 1989.